Loss Functions as Noise Models

Signal Detection and Estimation Theory, Project 3

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The problem

- Each loss function is the negative log-likelihood of one noise model.
- MSE assumes Gaussian noise. MAE assumes Laplace noise.
- **Question:** what is the cost when the noise model is wrong?

Model. We estimate a constant A from N samples:

$$x[n] = A + w[n], \qquad n = 0, \ldots, N-1, \qquad w[n] \text{ i.i.d., zero mean.}$$

This is linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \mathbf{w}$ with $\mathbf{X} = \mathbf{1}$.

Measure of cost.

$$\text{efficiency ratio} = \frac{\text{Var}(\hat{A})}{\text{CRLB}}$$

A ratio of 2 means twice the variance of the best estimator. If both variances fall as 1/N, the estimator needs 2 times more data. If the variance stops falling, more data will not help.

The loss comes from the noise model

Gaussian noise

$$\ln p(\mathbf{x}; A) = c - \frac{1}{2\sigma^2} \sum_n (x[n] - A)^2$$

The MLE minimises the squared error.

$$\Rightarrow \hat{A} = \bar{x} \quad \text{(mean)}$$

Laplace noise

$$\ln p(\mathbf{x}; A) = c - \frac{1}{b} \sum_n |x[n] - A|$$

The MLE minimises the absolute error.

$$\Rightarrow \hat{A} = \text{median}(\mathbf{x})$$

- For the median, the derivative of $\sum |x - A|$ is (samples below A) $-$ (samples above A). It changes sign at the middle sample.
- Choosing a loss is thus to choose a noise model!

CRLB

- Gaussian: σ^2/N
- Laplace: b^2/N

Variance of each estimator

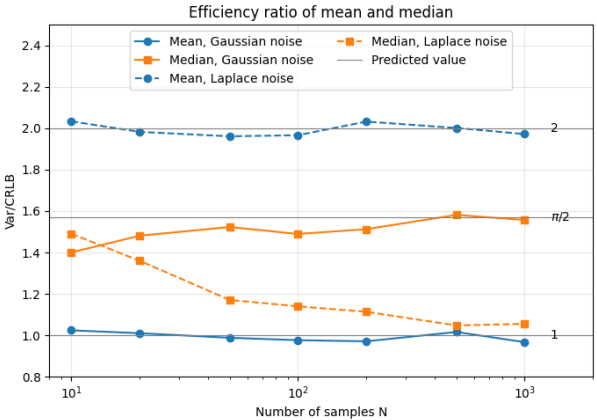
- Mean: $\mathrm{Var}(w)/N$ (exact for all N)
- Median: $\approx 1/(4N\,p_w(0)^2)$ (large N)

Predicted efficiency ratio

	Gaussian	Laplace
Mean (MSE)	1	2
Median (MAE)	$\pi/2$	1

- Each estimator is best only for its own noise.
- Under Laplace noise, the mean needs 2 times more data than the median.

Simulation with Gaussian and Laplace noise

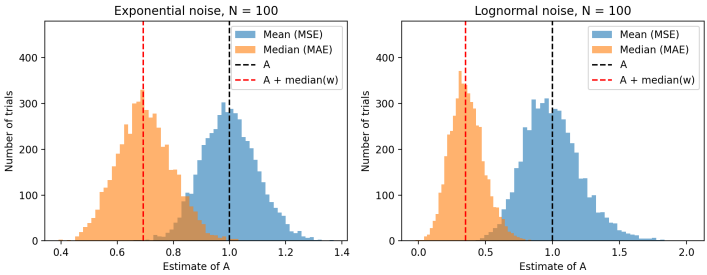


5000 trials. Measured at $N = 100$:

	Gauss.	Laplace
Mean	0.97	2.01
Median	1.51	1.17

- The mean is flat for all N.
- The median reaches its value only for large N.
- At small N, our median results match exact published values (NIST, 1964).

Skewed noise: the loss changes the target



Consider the case of zero-mean noise, but now not symmetric!

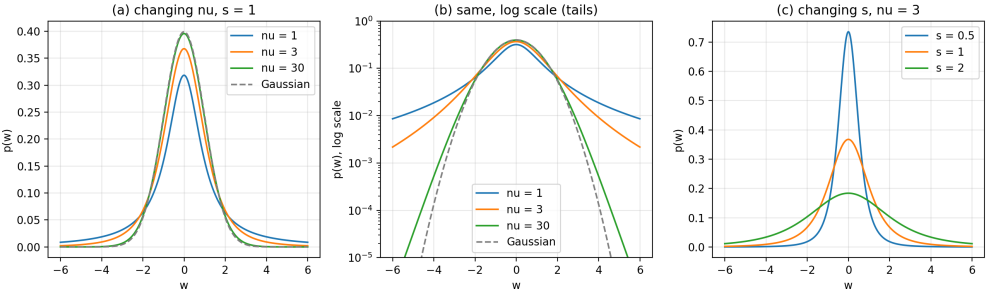
- Mean $= A$
- Median $= A + \text{median}(w)$

	Pred.	Meas.
Expon.	0.693	0.698
Lognorm.	0.351	0.363

MSE is not biased. MSE and MAE estimate different values.

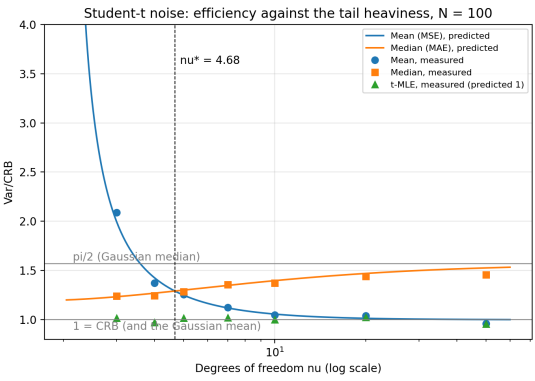
Student-t noise: the model

$$p_w(w) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)s}\left(1+\frac{w^2}{\nu s^2}\right)^{-\frac{\nu+1}{2}}$$



- ν sets the tails. $\nu \to \infty$ gives Gaussian. $\nu = 1$ gives Cauchy.
- s sets the width. $\mathrm{Var}(w) = s^2\nu/(\nu-2)$, finite only for $\nu > 2$.
- It is a Gaussian with a random variance (Lange et al., 1989).

Student-t noise: results



$$\mathrm{CRLB} = \frac{\nu+3}{\nu+1} \cdot \frac{s^2}{N}$$

$$\frac{\mathrm{Var}(\bar{x})}{\mathrm{CRLB}} = \frac{\nu(\nu+1)}{(\nu-2)(\nu+3)}$$

- Heavy tails make the mean much worse (leftwards in the plot).
- The median stays between 1.2 and 1.5.
- They are equal at $\nu \approx 4.68$.
- The t-MLE is a weighted mean with weight $(\nu+1)/(\nu+r^2)$. It reaches the CRLB.

Why we need a new bound

- The CRLB is computed from the **true** noise. An estimator built on the **wrong** noise model never reaches it.
- The mean under Laplace noise stays at $2\times$ CRLB for every N. The median under skewed noise does not even aim at A.
- We want a bound that predicts what the wrong-model estimator **actually** gets.

Example. Estimate the noise variance with a Gaussian model, $\hat{\theta} = \frac{1}{N}\sum_n x[n]^2$, when the noise is Laplace(b):

Bound	$N\cdot$ value
CRLB of the assumed Gaussian model	$8b^4$
CRLB of the true Laplace model	$16b^4$
Actual variance of $\hat{\theta}$ (measured 20.04)	$\mathbf{20b^4}$

Trusting the wrong model gives $8b^4$, which is not even a valid bound. The true CRLB is valid, but loose.

The misspecified CRB (MCRB)

True density p, assumed model $f(x \mid \theta)$.

Pseudo-true parameter (closest model in KL)

$$\theta_0 = \arg\max_{\theta} E_p\big[\ln f(x \mid \theta)\big]$$

MCRB (Vuong 1986, Fortunati et al. 2017)

$$\mathrm{MCRB} = \frac{1}{N}\frac{\mathcal{B}}{\mathcal{A}^2}, \qquad \mathcal{A} = E_p[\partial_\theta^2 \ln f] \qquad \mathcal{B} = E_p[(\partial_\theta \ln f)^2]$$

Model inside, true density outside.

Two key facts

- Correct model: $\mathcal{B} = -\mathcal{A} = I$, so MCRB = CRLB.
- The wrong-model MLE goes to θ_0 with variance $\to$ MCRB. So the bound is **tight**.

Our earlier formulas are MCRBs

- Gaussian model: mean, $\mathrm{Var}(w)/N$
- Laplace model: median, $1/(4Nf(m)^2)$, aiming at $A + \mathrm{median}(w)$

Efficiency ratio = MCRB / CRLB = the price of the wrong model.

Which estimators does the MCRB bound?

Paradox. Laplace noise, Gaussian model: MCRB $= 2b^2/N$. The median is unbiased for $\theta_0 = A$, yet has variance b^2/N.

Resolution. The bound needs $E_p[\hat{\theta}] = \theta_0(p)$ for **every** true p, not only ours.

- Perturb the truth: $p_t = p\,(1 + t\,g)$. Unbiasedness along p_t plus Cauchy–Schwarz gives

$$\mathrm{Var}(\hat{\theta}) \geq \frac{(E[\phi\, g])^2}{N\, E[g^2]}, \qquad \phi = -\frac{\partial_\theta \ln f}{\mathcal{A}}.$$

- The best direction is $g = \phi$, which gives exactly $\mathcal{B}/(N\mathcal{A}^2)$.
- Along that direction, θ_0 moves at rate $2b^2$ but the median moves at b^2. So the median is **not covered**.

So: one loss's MCRB is not a floor for other losses. The common floor is the CRLB of the true noise.

Regression with robust losses: one formula

$y_i = \theta_0 + \theta_1 x_i + w_i, \quad x_i \sim U(-2, 2), \quad$ loss ρ with $\psi = \rho'$.

Linearise $\sum_i \psi(y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\theta}})\, \mathbf{x}_i = \mathbf{0}$:

$$\mathrm{MCRB} = \frac{\kappa}{\lambda^2} (\mathbf{X}^\top \mathbf{X})^{-1}, \qquad \mathrm{CRLB} = \frac{(\mathbf{X}^\top \mathbf{X})^{-1}}{I_{\mathrm{loc}}}$$

$\lambda = E[\psi'(w)]$, $\kappa = E[\psi(w)^2]$ (MAE: $\lambda = 2p(0)$).

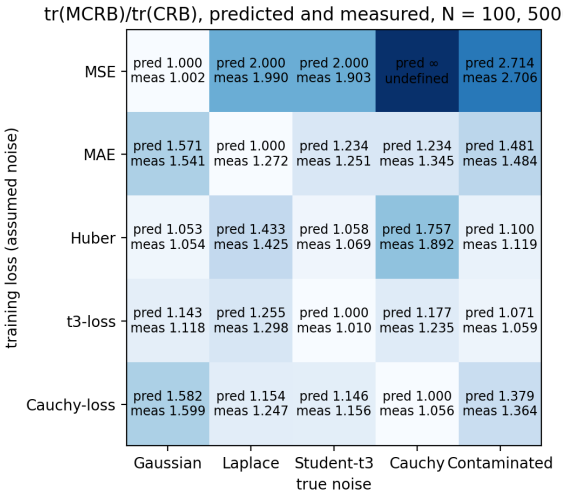
$$\text{ratio} = \frac{\kappa\, I_{\mathrm{loc}}}{\lambda^2} \;\geq\; 1$$

By Cauchy–Schwarz, with $= 1$ only for the true likelihood.

Losses: MSE, MAE, Huber, t_3, Cauchy.
Noises: Gaussian, Laplace, t_3, Cauchy, contaminated $0.9\,\mathcal{N}(0,1) + 0.1\,\mathcal{N}(0,25)$.

- Symmetric noise: every loss aims at the true $\boldsymbol{\theta}$.
- Scale rule fixed beforehand: each loss uses the scale that makes it the MLE for its own noise, estimated from the MAD.
- The ratio does not depend on $\mathbf{X}$.

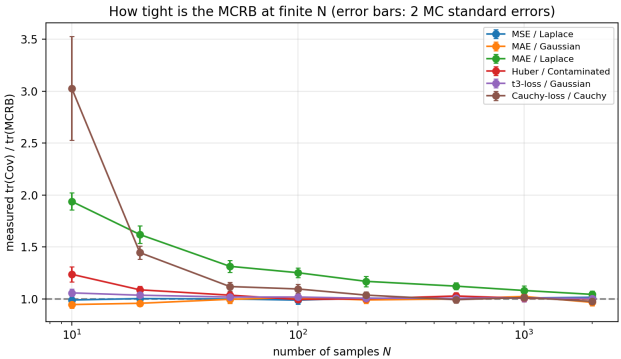
Mismatch matrix: predicted vs measured



$N = 100$, 5000 trials per cell.

- Diagonal $= 1$. No loss is best everywhere.
- MSE fails under Cauchy noise: $\kappa = E[w^2] = \infty$.
- Gates: MSE $\approx 1, 2$ under Gaussian and Laplace, t_3-loss at 1.01.
- 16 of 24 cells within 2 standard errors. All biases < 0.005.
- The rest are above the prediction: finite-N effects. At $N = 1000$ they are within 4%.

How tight is the MCRB at finite N?

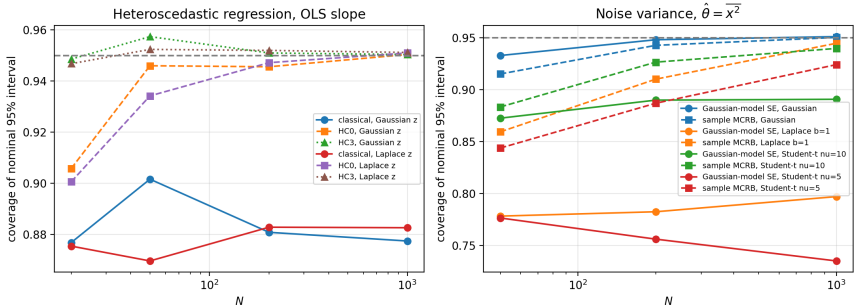


Measured variance / MCRB, 4000 trials.

- **Exact for all N:** MSE (OLS is linear in the data).
- **Slowly from above:** MAE under Laplace, $1.94 \to 1.04$. Cauchy-loss, 3.0 at $N = 10$.
- **Below the bound:** MAE under Gaussian, 0.942 ± 0.006 at $N = 10$.

The MCRB is reached in the limit. It is not a finite-N floor.

Estimating the MCRB from data (sandwich)



$\widehat{\mathrm{Cov}} = \hat{\mathbf{A}}^{-1}\hat{\mathbf{B}}\hat{\mathbf{A}}^{-1}$ from the residuals. Coverage of 95% intervals, 5000 trials:

- **Heteroscedastic regression:** classical SE about 88% at every N. HC3 (sandwich) about 95% from $N = 20$.
- **Noise variance, Laplace:** Gaussian-model SE 78–80% (predicted 78.5%). Sandwich reaches 94.5% at $N = 1000$.

Summary of part 2

- The CRLB says what is possible. The MCRB says what the wrong model actually gets.
- MCRB / CRLB is the price of the wrong loss, and one formula $\kappa\lambda^{-2}(\mathbf{X}^\top\mathbf{X})^{-1}$ predicts all 25 cells.
- It binds only estimators unbiased for $\theta_0(p)$ for every p. Compare losses against the true CRLB.
- Tight as $N \to \infty$. At small N it can be above or below.
- From data, the sandwich gives correct intervals where the model-based formula does not.

Thank you. Questions?